

US Patent & Trademark Office

Patent Public Search | Text View

United States Patent	12404731
Kind Code	B2
Date of Patent	September 02, 2025
Inventor(s)	Fosler; Matthew et al.

Drilling tool changer apparatus

Abstract

An apparatus for changing a drilling tool for a drilling rig is described, where the drilling rig comprises a drill tower supporting a drill pipe. The apparatus for changing a drilling tool has a drilling tool changer assembly moveable between a storage position and an exchange position, and the drilling tool changer assembly supports a rotatable carousel assembly. The rotatable carousel assembly has a plurality of bit adaptors for holding drilling tools and is selectively rotatable to bring a selected drilling tool into coaxial alignment with the drill pipe when the drilling tool changer assembly supporting the rotatable carousel assembly is moved into the exchange position. The apparatus includes mechanisms for aligning the selected drilling tool within a bit adaptor holding the selected drilling tool. The apparatus further includes mechanisms for aligning the drill pipe or a hammer with the selected drilling tool.

Inventors:	Fosler; Matthew (Wylie, TX), Tweedie; Steven B. (Plano, TX)
Applicant:	Epiroc Drilling Solutions, LLC (Garland, TX)
Family ID:	91667376
Assignee:	Epiroc Drilling Solutions, LLC (Garland, TX)
Appl. No.:	18/406550
Filed:	January 08, 2024

Prior Publication Data

Document Identifier	Publication Date
US 20240218752 A1	Jul. 04, 2024

Related U.S. Application Data

continuation parent-doc US 17161880 20210129 US 11466522 20221011 child-doc US 17899012
continuation-in-part parent-doc US 17899012 20220830 US 11898406 child-doc US 18406550

Publication Classification

Int. Cl.: E21B19/14 (20060101); E21B19/20 (20060101)

U.S. Cl.:

CPC E21B19/20 (20130101); E21B19/146 (20130101);

Field of Classification Search

CPC: E21B (19/146); E21B (19/20)

References Cited

U.S. PATENT DOCUMENTS

Patent No.	Issued Date	Patentee Name	U.S. Cl.	CPC
6457526	12/2001	Dailey	166/356	E21B 21/001
8342236	12/2012	Bienfang	166/77.51	E21B 19/18
11220872	12/2021	Diekmann	N/A	E21B 3/02
11466522	12/2021	Fosler	N/A	E21B 19/20
11898406	12/2023	Fosler	N/A	E21B 19/146
2014/0083774	12/2013	Hoult	175/52	E21B 17/00
2014/0338973	12/2013	Taylor	175/24	E21B 19/146
2018/0291696	12/2017	Amerson	N/A	E21B 19/18
2022/0018196	12/2021	Gregg	N/A	B63B 35/4413

FOREIGN PATENT DOCUMENTS

Patent No.	Application Date	Country	CPC
WO-2019226096	12/2018	WO	E21B 1/00

Primary Examiner: Buck; Matthew R

Attorney, Agent or Firm: Harper & Bates LLP

Background/Summary

CROSS-REFERENCE TO RELATED APPLICATIONS (1) This application is a continuation-in-part of U.S. Non-Provisional patent application Ser. No. 17/899,012 filed Aug. 30, 2022 and entitled “Drilling Tool Changer Apparatus”, now U.S. Pat. No. 11,898,406, which is a continuation of U.S. Non-Provisional patent application Ser. No. 17/161,880 filed Jan. 29, 2021 and entitled “Drilling Tool Changer Apparatus”, now U.S. Pat. No. 11,466,522, each of which is incorporated herein by reference for all purposes.

TECHNICAL FIELD

(1) This disclosure relates to drilling systems and more particularly to a drilling tool changer apparatus for use in applications such as mining, blast-hole drilling and other down-hole drilling applications.

BACKGROUND

(2) This background section provides a context to the invention recited in the claims. The

description here may include concepts that could be pursued but are not necessarily ones that have been previously conceived or pursued. Therefore, unless otherwise indicated, what is described in this section is not prior art to the description and claims in this application and is not admitted to be prior art by inclusion in this section.

(3) Drilling systems generally have a vertical drill tower, or mast, constructed from structural members such as steel beams and reinforcing supports. The drill tower is often coupled to a mobile platform, usually self-propelled, for positioning the drill tower in a desired location to conduct a drilling operation. The drill tower is often equipped with a drill magazine or holder of some sort, which is adapted to support a drill string formed from a combination of drill extenders, usually called drill rods or drill pipes. The drill magazine is used to selectively add the drill pipes to the drill string for drilling a hole having a desired depth. The drill magazine is intended to allow a drilling operation to progress into the drill hole by making readily available a continuous string of drill pipes as needed for advancing a drilling tool into a drill hole.

(4) The lowermost drill pipe in the drill string is configured to receive a drilling tool at its lower end to conduct the drilling operation. The drilling tool is usually a drill bit or a down-the-hole (“DTH”) bit and hammer chuck assembly but could be a sub with a drilling tool attached. (In this disclosure, the term “drill bit” may be used interchangeably for the general term “drilling tool”, and the term “drill pipe” may be used interchangeably for the term “drill string”, unless the context requires otherwise.) After a certain amount of usage in the drilling operation it is often desirable or necessary to remove and replace the drilling tool due to accumulated wear of an in-service drill bit, or the need to change between a drill bit and a hammering tool (or vice versa), etc. In order to minimize downtime in the drilling operation due to change-out of drilling tools, mechanisms may be provided to facilitate removing one drilling tool from the end of the drilling string and replacing it with another drilling tool from a storage or supply location.

(5) What is needed is a replaceable magazine or carousel for drilling tools that does not require a large amount of space to accommodate swinging of an arm for the magazine, or for rotation of the magazine, which magazine can accommodate selecting drilling tools other than in a linear order, and which is readily adaptable for use in an automated or semi-automated manner to improve access for changing drilling tools, to minimize downtime during change-outs and to minimize the number of personnel required at the drilling rig.

SUMMARY

(6) In accordance with one embodiment, an apparatus for changing a drilling tool for a drilling rig is described, where the drilling rig comprises a drill tower supporting a drill pipe. The apparatus for changing a drilling tool comprises a drilling tool changer assembly moveable between a storage position and an exchange position; where the drilling tool changer assembly supports a rotatable carousel assembly. The rotatable carousel assembly has a plurality of bit adaptors for holding drilling tools and is selectively rotatable to bring a selected drilling tool into coaxial alignment with the drill pipe when the drilling tool changer assembly supporting the rotatable carousel assembly is moved into the exchange position. The apparatus may include means for aligning the selected drilling tool within a bit adaptor holding the selected drilling tool, and/or the apparatus may include means for aligning the drill pipe or a hammer with the selected drilling tool.

(7) This Summary is provided to introduce a selection of concepts in a simplified form that are further described below in the Detailed Description. This Summary is not intended to identify key features or essential features of the claimed subject matter, nor is this Summary intended to be used to limit the scope of the claimed subject matter.

Description

BRIEF DESCRIPTION OF THE DRAWINGS

- (1) Non-limiting embodiments of the present disclosure are described by way of example in the following drawings, which are schematic and are not intended to be drawn to scale:
- (2) FIG. 1 shows a side view of an embodiment of a typical drilling machine having a tower capable of tilting, where the tower is raised to a vertical position.
- (3) FIG. 2 shows schematically a general control system of the drilling machine, where a computer and peripherals are connected to operate the disclosed drilling tool changer apparatus.
- (4) FIG. 3 is a perspective view of the rig platform showing a drilling tool changer assembly positioned near the break-out area of the tower.
- (5) FIG. 4 is a side view of the structure in FIG. 3 showing the lid to the drilling tool changer assembly enclosure opened.
- (6) FIG. 5 is a side view of the structure of FIG. 4, showing the drilling tool changer assembly tilted to engage with a drill pipe in the tower; the drill pipe having a drilling tool attached.
- (7) FIG. 6 is a side view of the structure of FIG. 4, showing the drilling tool changer assembly tilted and engaged with a drill pipe for removal or addition of a drilling tool to the drill pipe.
- (8) FIG. 7 is a side view of the structure of FIG. 4 showing the drilling tool changer assembly tilted and a drill pipe withdrawn from the assembly after removal of a drilling tool from the drill pipe.
- (9) FIG. 8 is a perspective view of the first embodiment of the carousel of a drilling tool changer assembly including an enclosure, with bit adaptors shown holding a plurality of drilling tools.
- (10) FIGS. 9A and 9B comprise perspective views of an embodiment of the carousel of the drilling tool changer assembly, where FIG. 9A is a top view of the assembled carousel shown holding drill bits, and FIG. 9B is a perspective view of an embodiment of the rotatable carousel assembly viewed from the rear.
- (11) FIGS. 10A and 10B comprise views of an alignment plate of the rotatable carousel of the drilling tool changer assembly, where FIG. 10A is a perspective view of the alignment plate showing alignment pins to be received by the carousel plate, and FIG. 10B is a plan view of the structure of FIG. 10A, showing the off-center location of one alignment pin.
- (12) FIG. 11 is a perspective view of an embodiment of the drilling tool changer assembly, viewed from the underside of its enclosure and illustrating a sensor arrangement for detecting the angular position of the rotatable carousel.
- (13) FIGS. 12A and 12B comprise views of the rotatable carousel of the drilling tool changer assembly, where FIG. 12A is a plan view of the rotatable carousel of the drilling tool changer assembly, and FIG. 12B is a perspective view of the carousel without drilling tools in the holders for the same.
- (14) FIGS. 13A and 13B comprise perspective views of bit adaptors; FIG. 13A is an adaptor having a welded bottom plate, and FIG. 13B is an adaptor having a removable bottom plate.
- (15) FIGS. 14A and 14B comprise perspective views of an embodiment of the bit adaptors for the drilling tool changer. FIG. 14A shows empty bit holders on the carousel of the drilling tool changer assembly; FIG. 14B shows a plurality of bits or drilling tools disposed in bit adaptors of possibly varying sizes on a carousel plate, and FIG. 14B shows a typical bit adaptor.
- (16) FIGS. 15A and 15B comprise views of an embodiment of the drilling tool changer assembly, where FIG. 15A is a plan view depicting bit adaptors of various sizes, and FIG. 15B is a side cut-away view of bit adaptors of various sizes on a carousel.
- (17) FIG. 16A is a side plan view of a down-the-hole (DTH) tool positioned within a bit adaptor, and FIG. 16B depicts a side plan view of a DTH tool positioned within an embodiment of a pivoting bit adaptor.
- (18) FIG. 17A is a top-down perspective view of the pivoting bit adaptor of FIG. 16B in the open position, and FIG. 17B is another top-down perspective view of the pivoting bit adaptor of FIG. 16B in the open position and rotated approximately 90 degrees.

(19) FIGS. **18A**, **18B** and **18C** comprise a sequence of side cross-sectional views of a DTH tool being lowered into the pivoting bit adaptor of FIG. **16B**, FIG. **17A** and FIG. **17B**, where FIG. **18A** shows the pivoting bit adaptor in the open position with the DTH tool making initial contact with a shim pad on a pivoting component, FIG. **18B** shows the pivoting bit adaptor with the pivoting component in a partially rotated position with the DTH tool partially lowered into the pivoting bit adaptor, and FIG. **18C** shows the pivoting bit adaptor with the pivoting component in a fully rotated position with the DTH tool fully lowered into the pivoting bit adaptor.

(20) FIGS. **19A** and **19B** comprise perspective views of an embodiment of a side swinging drill string support, where FIG. **19A** shows the side swinging drill string support in a stowed configuration, and FIG. **19B** shows the side swinging drill string support in an activated configuration.

(21) FIG. **20A** is a perspective view of an embodiment of a side swinging drill string support with a roller sized for a larger diameter drill string. FIG. **20B** is a perspective view of another embodiment of a side swinging drill string support with a roller sized for a smaller diameter drill string.

(22) FIGS. **21A** and **21B** comprise a sequence of side, partial cut-away views of a side swinging drill string support in a representative operational environment, where FIG. **21A** shows the side swinging drill string support in the stowed configuration and a hammer or drill pipe on the end of a drill string sagging out of alignment, and FIG. **21B** shows the side swinging drill string support in the activated configuration with its roller engaging the hammer or drill pipe to align it with the drilling tool.

(23) FIGS. **22A** and **22B** comprise perspective views of an embodiment of an alignment clamp, where FIG. **22A** shows the alignment clamp in an open position, and FIG. **22B** shows the alignment clamp in a closed position.

(24) FIGS. **23A**, **23B** and **23C** comprise a sequence of top plan views of the alignment clamp of FIGS. **22A** and **22B** in a representative operational environment, where FIG. **23A** shows the alignment clamp in an open position and a DTH tool in a tilted position within a bit adaptor, FIG. **23B** shows the alignment clamp in a partially closed position, making initial contact with the DTH tool, and FIG. **23C** shows the alignment clamp in a closed position with the DTH tool aligned with the bit adaptor.

(25) FIG. **24** is a side perspective view of the structure of FIG. **23C**, with the alignment clamp shown in the closed position and with the DTH tool aligned with the bit adaptor.

(26) FIGS. **25A**, **25B**, **25C**, **25D** and **25E** comprise a sequence of side perspective views of an embodiment of an enclosure that houses lower and upper alignment clamps in a representative operational environment. FIG. **25A** shows the lower and upper alignment clamps in open positions, respectively, and a DTH tool tilted within a bit adaptor carried by a carousel assembly within the enclosure. FIG. **25B** shows the lower alignment clamp in a closed position around the DTH tool to align the bit within the bit adaptor. FIG. **25C** shows the upper alignment clamp in a closed position to guide a hammer on the end of a drill string into alignment with the DTH tool. FIG. **25D** shows the lower and upper alignment clamps in closed positions, respectively, to maintain alignment as the hammer is threaded onto the DTH tool. FIG. **25E** shows the lower and upper alignment clamps retracted to open positions, respectively, once the joint is made. FIG. **25F** shows the lower and upper alignment clamps in open positions, respectively, and the drill string, hammer, and DTH tool retracted out of the enclosure.

DETAILED DESCRIPTION

(27) FIG. **1** shows a side view of a typical mobile drilling machine **100**. The drilling machine has a tower **110**, a platform **120** supporting the tower **110**, and typically tracks **130** or wheels for propelling the drilling machine over ground. FIG. **1** also shows an operator's cab **140** situated on the platform **120**. The term “cab” in this disclosure refers to either a housing for an operator or a workstation location on the platform **120**, which may or may not be occupied by an operator; the latter would be the case in autonomous machines. FIG. **1** shows the typical engine compartment or

“power pack” **105**, and mechanical accessories of a drilling machine **100**. The tower **110** will also carry a drill string **150** extending through the tower **110**, and a rotary head **145**, being typically a hydraulic or electric motor for rotating the drill string **150**. The drill string **150** is terminated at its lower end by a removable bit or drilling tool **215**. In general, the tower **110** is pivotably moveable so that its tilt allows operations for the change-out of drilling tools **215**. The drill string **150** and its components will be discussed in more detail below. The reader should note that the improvements disclosed here could be embodied in a fixed drilling machine as well as the mobile machine illustrated, and the claims cover both embodiments.

(28) The drilling machine **100** further includes a control system **155**, which is operatively coupled to the power pack **105**. The associated implements operatively connected to the power pack **105** and the control system **155** are described in more detail below and suggested only schematically in FIG. 1. FIG. 2 shows schematically the control system **155** operatively associated with the example drilling machine which here includes the disclosed drilling tool changer apparatus. The control system **155** comprises generally a computer **160** that is typically a programmable digital computer, further comprising a read-only memory, a non-transitory computer readable storage medium for storing instructions executable by a processor (such as a random-access memory), a central-processing unit or processor, and a hard drive or flash memory or the like for further storage of programs and data, as well as input and output ports. FIG. 2 shows the example control system **155** operatively connecting the computer **160** to a visual display **165** for an operator, and a control input interface **170**, such as a joystick, or touch screen, or both. In this disclosure, the term “computer” may be used broadly to refer to both a programmable digital computer as just described, the software executed by the computer, and relevant peripheral devices connected to a computer, as well as networks of computers. One industrial example of a computer **160** suitable for controlling the operations of actuators in a tracked vehicle is the Rig Control System (RCS) provided by Epiroc Drilling Solutions, LLC. The computer **160** connects by means of a bus **115**, which may be a wired or wireless network, to the various sensors and actuators shown in FIG. 2 (and in some embodiments, others).

(29) The control system **155** includes one or more control inputs which can be adjusted by the operator in the operator's cab **140**. These may include one or more input controls for controlling the operation of the tower **110**, including its tilt angle, and also the operations of the drilling tool changer assembly **190**, described below, which is a component of the drilling tool changer apparatus.

(30) FIG. 3 is a perspective view of the rig platform showing the drilling tool changer assembly **190** positioned near the break-out area of the tower. Referring to FIG. 3, we see a drilling tool changer assembly **190**, comprising an enclosure **195** for the carousel assembly **230** (see FIGS. 7 and 8). The drilling tool changer assembly **190** is disposed on the rig platform **120** and mounted by a hinge **220** so as to rotate upward from the rear (to the left in FIG. 3) when urged upward by changer actuators **225**. FIG. 3 also shows break-out tool **180** adjacent to the base of the tower **110**, and the operator's cab **140** that would typically be situated adjacent to tower **110** on the drilling rig **100**. The enclosure **195** has a lid **200** that can be opened by action of one or more lid actuators **210**, which may be hydraulic or electric motors or the like. The lid **200** is not required, but is desirable to keep dirt and foreign objects from falling into the drilling tool changer assembly **190**.

(31) It is advantageous that the disclosed drilling tool changer assembly **190** is not located over the opening in the drill deck **120**, where it could interfere with breakout operations, and further, is not located on or connected to the tower **110**, where it would add extra weight and complicate balancing the tower **110**.

(32) FIG. 4 shows a side view of the structures of FIG. 3, where the lid **200** of the carousel enclosure **195** is opened. In FIG. 5, the drilling tool changer assembly **190**, with the lid **200** of the carousel enclosure **195** opened, is tilted forward on the hinge **220** by action of actuators **225**, thus moving from a storage position as shown in FIGS. 3 and 4, to an exchange position, as shown in

FIGS. 5 and 6. FIG. 6 shows the tilted drilling tool changer assembly **190** receiving a drilling tool **215** from the drill string **150** held by the now-tilted tower **110**, where the drilling tool changer assembly **190** is tilted at an angle to substantially match the tilt of the tower **110**. FIG. 7 shows a drill string **150** after a drilling tool **215** has been removed and placed in the drilling tool changer assembly **190**. FIGS. 5-7 could also illustrate the opposite operation, where a drilling tool **215** is attached to the drill string **150**. FIGS. 5-7 show an actuator or motor **280** for rotating the carousel assembly **230**. (The actuator or motor **280** may include a gearbox.) The apparatus and procedures for the change-out or replacement of the drilling tool **215** are described in more detail below.

Drilling Tool Changer Carousel Assembly

(33) FIG. 8 is a perspective view of an embodiment of the rotatable carousel assembly **230** of a drilling tool changer assembly **190**, situated in an enclosure **195**, with bit adaptors **235** shown holding a plurality of drilling tools **215**. The rotatable carousel assembly **230** is removable from the drilling tool changer assembly **190** and replaceable therein, as will be described. The rotatable carousel assembly **230** has a plurality of receptacles, called here bit adaptors **235**, for receiving drilling tools **215**. The rotatable carousel assembly **230** is selectively rotatable, in either direction, via a motor **280**, to bring a selected drilling tool **215** into coaxial alignment with the drill string **150** when the drilling tool changer assembly **190** supporting the rotatable carousel assembly **230** is moved into the exchange position as shown in FIGS. 5-7. FIG. 8 also shows an empty bit adaptor **235** in a pre-determined angular position **420** of the rotatable carousel assembly **230**, where a used drilling tool **215** may be deposited after removal from the drill string **150**. Alternatively, this pre-determined angular position **420** could be the current position of a drilling tool **215** held in the bit adaptor **235**, to be attached to the drill string **150**. In either case, the rotatable carousel assembly **230** will be rotated to bring the relevant bit adaptor **235** into the correct pre-determined angular position **420** for the pick-up or drop-off of a drilling tool **215** as determined by an operator or the control system **155**.

(34) The rotatable carousel assembly **230** has at least one alignment pin **245** situated on the alignment plate **255** below a carousel plate **285**, which alignment pin **245** passes through a slot, notch, or hole **290** in the carousel plate **285**, so that the carousel plate **285** can be held on the alignment plate **255** in a fixed angular position within the rotatable carousel assembly **230**. The carousel plate **285** of this embodiment further comprises at least one off-center alignment hole **295**, comprising a slot, notch or hole, mating with an off-center alignment pin **250**, for re-aligning the rotatable carousel assembly **230** in the same angular position on the alignment plate **255** after the rotatable carousel assembly **230** is removed from the drilling tool changer assembly **190** and replaced in the drilling tool changer assembly **190**. In this way the rig operator, or an automatic process in a control system **155**, can know which drilling tool **215** is in which bit adaptor **235** in the rotatable carousel assembly **230**. The feature allows the operator or control system **155** to confirm or select the correct drilling tool **215** for the desired drilling task, because in general, drilling tools **215** may be of different sizes, or types, or even have different conditions of wear, and it is highly advantageous to know without manual inspection which drilling tool **215** is in which position in the rotatable carousel assembly **230**, so that drilling operations are not delayed after a change-out of drilling tools **215**.

(35) The reader should note that the alignment pin or pins **245**, and the off-center alignment pin **250** are shown schematically in the figures, and represent, in general, an indexing feature. These may be any sort of finger, rod, or lug that can engage slots, notches or holes in the carousel plate **285** to maintain the same in a given angular position in the rotatable carousel assembly **230**, and thus the terms “alignment pin” or “off-center alignment pin” should be construed broadly according to this indexing function. In particular, in some embodiments therefore, alignment pins **245** or off-center alignment pin **250** do not pierce the carousel plate **285** completely in order to engage it.

(36) In some embodiments, the rotatable carousel assembly **230** may optionally be a first rotatable

carousel assembly **230**, and the rotatable carousel assembly **230** replaced in the drilling tool changer assembly **190** may be a second rotatable carousel assembly **230**, the second rotatable carousel **230** having possibly replaced or exchanged drilling tools **215**.

(37) FIG. **9A** shows examples of drilling tools **215** resting in the bit adaptors **235**. Each bit adaptor **235** is connected to a carousel plate **285**. FIG. **9A** shows alignment holes **290** in the carousel plate **285** for selectively receiving an alignment pin **245**, and also shows at least one off-center alignment hole **295** and off-center alignment pin **250**, to allow indexing of the carousel plate **285**. The carousel plate **285** may be a single plate, or constructed of two plates bolted together by bolts or retaining pins **240**, for ease of manufacturing.

(38) FIG. **9A** shows the rotatable carousel assembly **230** as described in above, but including the alignment plate **255** that supports the alignment pins **245** and off-center alignment pin or pins **250**. (Generally, only one off-center pin **250** would be required for alignment, but more than one could be provided, and reference here to an “alignment pin” includes one or more.) The carousel plate **285** and the bit adaptors **235** are preferably bolted together with bolts **240**. FIG. **9A** shows the alignment plate **255** supporting the alignment pins **245** and off-center alignment pin **250**, rotatable by a bull gear **260**. As shown in FIG. **9A**, the carousel plate **285** is removably supported by the alignment plate **255**, and the alignment pins **245** and off-center alignment pin **250** thus pass through holes or slots **290** in the carousel plate **285**. One of the holes or slots is an off-center alignment hole **295**, so that the carousel plate **285** (and thus the bit adaptors **235**) can only be indexed to and replaced in one position on the alignment plate **255**.

(39) FIG. **9B** is a perspective view of this embodiment of the rotatable carousel assembly **230** viewed from the rear or bottom of the assembly depicted in FIG. **9A**. FIG. **9B** shows how the carousel plate **285** is supported on the alignment plate **255**, and how the alignment plate **255** is fastened to a bull gear **260**, which bull gear **260** rotates the alignment plate **255**, and thus the carousel plate **285**, when rotation is imparted to the bull gear **260** by an actuator, such as a motor **280** and pinion gear (not shown in FIG. **9B**). The rotation imparted may be in either direction, under command of an operator or an automatic process in a control system **155**, to index the rotatable carousel assembly **230** to the desired position, as explained below.

(40) FIG. **10A** show a perspective view of the alignment plate **255** supported by the bull gear **260**. The alignment plate **255** supports a plurality of alignment pins **245**, and at least one off-center alignment pin **250**. As described above with reference to FIGS. **9A** and **9B**, the carousel plate **285** is supported by the alignment plate **255**, where alignment holes **290** (and off-center alignment hole **295**) in the carousel plate **285** receive, respectively, alignment pins **245** and the at-least one off-center alignment pin **250**. This structure is shown in plan view in FIG. **10B**, where the circumference **300** of a circle on the radius of the alignment pins **245** is shown, illustrating the off-center location of the at least one off-center alignment pin **250**, as well as the respective alignment holes **290** and off-center hole **295** in the carousel plate.

(41) FIG. **11** illustrates an embodiment of a sensor arrangement for detecting the angular position of the rotatable carousel assembly **230**. A magnet **370** or equivalent is connected to a pin or shaft **390**, which pin or shaft **390** is connected to the alignment plate **255** and compelled to rotate therewith. A sensor **380** detecting the position of the magnet **370** is fixed to a bracket **400** or other convenient structure that is fixed to the enclosure **195** of the drilling tool changer assembly **190**, so that as the alignment plate **255** rotates by action of the motor **280**, the change in output from the sensor **380** corresponds to the angular position of the alignment plate **255**, and thus the angular position of the rotatable carousel assembly **230**, which is rotated by action of the alignment plate **255**. Because the rotatable carousel assembly **230** only fits the alignment plate **255** in one pre-determined position, the output of the sensor **380** uniquely reports the angular position of the rotatable carousel assembly **230**, which output is preferably communicated to the control system **155** by means of a connector **410**, which can be a cable or wireless connection. A suitable magnetic sensor of this type would be the SMART Position Sensor configuration, manufactured by

Honeywell International, Inc., or equivalent.

(42) FIGS. 12A and 12B are a more detailed view of the carousel assembly 230 of the drilling tool changer assembly 190, FIG. 12A being a plan view and FIG. 12B being a perspective view of the same. In FIG. 12A, three bit adaptors 235 are occupied by drilling tools 215, and one bit adaptor 235 is empty. (The reader should note that the number of bit adaptors 235 in a carousel assembly may be more or fewer than the four shown.) In FIG. 12B, for clarity, all the bit adaptors 235 are shown as empty.

(43) FIGS. 12A and 12B show a carousel plate 285, which may be one integral plate, or, for convenience of construction, two or more plates fastened together, with bolts, such as the retaining pins or bolts 240 shown. The carousel plate 285 supports bit adaptors 235 (described in more detail below), for receiving drilling tools 215. The carousel plate 285 has holes 290 for receiving alignment pins 245, and at least one off-center hole 295 for receiving an at least one off-center alignment pin 250. In this way, a particular type of drilling tool 215 located in a particular bit adaptor 235, and be replaced in the same position in the carousel assembly 230, if the carousel assembly 230 is removed and replaced in the drilling tool changer assembly 190.

(44) Further referring to FIGS. 12A and 12B, the figures show lifting rings 270, where by a lifting apparatus, such as a crane, can attach to and lift out the carousel assembly 230 for replacement of drilling tools 215 in the bit adaptors 235. The bit adaptors 235 are shown fastened to the carousel plate 285 by retaining pins or bolts 240. As illustrated below, each bit adaptor 235 further comprises a breaker plate 310 and a bit holder 320. Alternatively, the breaker plate 310 may be fastened to a bit holder plate 330 that may be removable. The breaker plate 310 of the bit adaptors 235 comprises nubs 275 in its inner circumference to hold the drilling tool 215 and resist the torque of the drill pipe 205 when the drill pipe 205 is rotated by the break-out tool mechanism 180, as a drilling tool 215 is added to or removed from the drill pipe 205. Therefore, no special sub or supplemental adaptor to the drilling tool 215 is required for the breakout operations.

(45) FIGS. 13A and 13B show more detail of bit adaptor 235 of the drilling tool changer assembly 190. In FIG. 13A, an embodiment has a breaker plate 310 with nubs 275, where the nubs 275 are sized to grasp a drilling tool 215 of a pre-determined size and restrain its movement against the torque of the break-out operation. In FIG. 13A, the bit holder 320 and the bit holder plate 330 are alternatively welded together to form an integral unit. Mounting holes 265 are provided for mounting the bit adaptor 235 to the carousel plate 285. In the embodiment of FIG. 13B, the breaker plate 310 is connected to a bit holder plate 330, where the bit holder plate 330 supports a drilling tool inserted into the bit adaptor 235. The embodiment of FIG. 13A, providing a more shallow receptacle, is adapted to support and hold smaller drilling tools 215 than the embodiment of FIG. 13B. In either embodiment, the bit adaptor functions to provide storage for the drilling tool 215 in the carousel assembly 230, while also holding the drilling tool 215 firmly against the torque of the break-out operation.

(46) FIGS. 14A and 14B show embodiments of the drilling tool changer assembly 190 where the bit adaptors 235 situated in the carousel assembly 230 are sized to receive drilling tools 215 of differing sizes, both in diameter or in depth, or both dimensions. FIG. 15A is a plan view of such an embodiment. FIG. 15B is a side view omitting depiction of the carousel plate 285 for clarity, and showing only the exemplary bit adaptors 235 of varying sizes. In FIG. 15B, the rightmost bit adaptor 235 depicted has a bit holder plate 330, but alternatively lacks a bit holder 320.

Detection of Drilling Tools in Changer Assembly

(47) It is advantageous for an operator or an automatic process to detect if a bit or drilling tool 215 is present in the rotatable carousel assembly 230, so that no attempt will be made to engage with the drill pipe 205 if no bit is present. Also, it is advantageous that an operator or automatic process can detect which type of bit or drilling tool 215 is actually present in a particular bit adaptor 235. To that end, embodiments can be provided with these capabilities. Radio-frequency identification (RFID) technology, for example, is known in the art. FIG. 8 illustrates an RFID sensor 340 situated

at some convenient location in the enclosure **195** of the rotatable carousel assembly **230**. The RFID sensor **340** can read a corresponding RFID tag (not shown) on a drilling tool **215** and transmit this identification to an operator or automatic process in the control system **155**, where it may be stored for further reference or use in the automatic process. For example, an automatic process could select the desired size and type of drilling tool **215** for particular rock strata or drilling conditions. Equivalently, electromagnetic sensors could be employed, such as near-field communications devices operating on principles generally within RFID technology, as is known in the art. In other embodiments, the RFID sensor **340** could be a plurality of RFID sensors **340**, where each is associated with a particular bit adaptor **235**.

(48) Further, FIG. **8** shows an ultrasonic detector **350**, which provides the capability of detecting whether or not a drilling tool **215** is present in a particular bit adaptor **235** of the rotatable carousel assembly **230**. In the case of both the RFID sensor **340** and the ultrasonic detector **350**, the relevant index or angular position of the rotatable carousel assembly **230** (as known by the means described in connection with FIG. **11** above) can be correlated with the identification returned from the RFID sensor **340**, or with the signal from the ultrasonic detector **350** to provide this information to an operator or to an automatic process in the control system **155**, where it may be stored for future reference or use in an automatic process. Equivalents in other embodiments could be a radar sensor to detect presence of a bit **215**, or a switch.

Down-the-hole (DTH) Tool and Hammer Alignment

(49) In some implementations, the drilling tool **215** may be a down-the-hole (“DTH”) drill bit and hammer chuck assembly (collectively referred to herein as a “DTH tool”), and the drill pipe **205** at the end of the drill string **150** may be replaced with a hammer that includes receiving threads to mate with the threads of the DTH tool. The DTH tool threads and the corresponding hammer threads are less tapered and therefore tend to be more susceptible to cross threading than the API or BECO threads on standard rotary drill bits and drill pipes. As such, a more precise alignment between the DTH tool and hammer may be required to make the joint and avoid cross-threading.

(50) However, because the DTH tool includes both a hammer bit and a hammer chuck, the DTH tool is relatively longer and more top heavy than standard rotary drill bits such that when the drilling tool changer assembly **190** is tilted to the exchange position shown in FIGS. **5-7**, the DTH tool is more likely to tilt within the bit adaptor **235** rather than staying aligned with the drill string **150**.

(51) FIG. **16A** depicts a side plan view of a DTH tool **515** positioned within the previously described bit adaptor **235** of the present disclosure, and FIG. **16B** depicts a side plan view of a DTH tool **515** positioned within an embodiment of a pivoting bit adaptor **535** of the present disclosure. As shown in FIG. **16A**, the top heavy DTH tool **515** tends to tilt over and fall out of alignment in the bit adaptor **235** when the drilling tool changer assembly **190** is in the exchange position, whereas the DTH tool **515** is held in an aligned position in the pivoting bit adaptor **535** when the drilling tool changer assembly **190** is in the exchange position.

(52) FIG. **17A** depicts a top-down perspective view of the pivoting bit adaptor **535** of the present disclosure, and FIG. **17B** depicts another top-down perspective view of the pivoting bit adaptor **535** rotated approximately 90 degrees. The pivoting bit adaptor **535** comprises a bit holder **520** with a bit holder plate **530** coupled to a bottom surface thereof to support a drilling tool **215**, such as the DTH tool **515** positioned within the pivoting bit adaptor **535**. A mounting ring **525** surrounds the bit holder **520** and includes mounting holes **565** for mounting the pivoting bit adaptor **535** to the carousel plate **285**. The pivoting bit adaptor **535** further comprises a pivoting component **510** that includes a semicircular support **508** and a shim support **512**. The shim support **512** is coupled to the semicircular support **508** and extends substantially perpendicular thereto. The shim support **512** forms a trough **513** into which a shim pad **514** is positioned. The shim pad **514** can be provided in different sizes for various types and sizes of bits.

(53) The pivoting component **510** is rotationally coupled to the bit holder **520** via a pin/flange

assembly **540**, and a return spring **545** ensures the pivoting component **510** remains in the open position shown in FIG. **17A** and FIG. **17B** when the pivoting bit adaptor **535** is empty. As best shown in FIG. **17B**, one or more keyed inserts **522** may be provided on an inside wall of the bit holder **520**, and these keyed inserts **522** can be provided in different sizes for various types and sizes of bits to inhibit rotation of the drilling tool **215**, such as the DTH tool **515**, when making or breaking a joint.

(54) FIG. **18A** through FIG. **18C** depict a sequence of side cross-sectional views of a DTH tool **515** being lowered into a pivoting bit adaptor **535** of the present disclosure. Although the pivoting bit adaptor **535** is shown in a horizontal position in this sequence, the pivoting bit adaptor **535** would perform in the same manner in a tilted position, such as the exchange position shown in FIGS. **5-7**.

(55) Referring to FIG. **18A**, the pivoting bit adaptor **535** is shown with the pivoting component **510** in the open position, and with the DTH tool **515** making initial contact with the shim pad **514**.

Referring to FIG. **18B**, as the DTH tool **515** continues to be lowered into the pivoting bit adaptor **535**, the DTH tool **515** pushes against the shim pad **514** until the force of the return spring **545** is overcome and the pivoting component **510** rotates about pin/flange assembly **540**. Referring to FIG. **18C**, when the DTH tool **515** is fully lowered into engagement with the bit holder plate **530**, the shim pad **514**, as well as the keyed insert(s) **522**, keep the DTH tool **515** snug within the pivoting bit adaptor **535**. The shim pad **514** and keyed insert(s) **522** are also sized to facilitate proper lowering of the pivoting component **510** to the position shown in FIG. **18C**. When the pivoting component **510** rotates to the position shown in FIG. **18C**, the semicircular support **508** extends horizontally/radially to surround a portion of the DTH tool **515** positioned within the pivoting bit adaptor **535**, and the shim support **512** extends vertically/axially within the pivoting bit adaptor **535**, with the shim pad **514** engaging the DTH tool **515**.

(56) Another alignment issue that may occur when the drill string **150** includes a hammer or a heavier drill pipe **205** at the end is sagging of the drill string **150** out of alignment with the drilling tool **215** when the tower **110** is tilted as shown in FIGS. **5-7**.

(57) Referring now to FIG. **19A** and FIG. **19B**, an embodiment of a side swinging drill string support **600** is depicted in a stowed configuration and in an activated configuration, respectively. The side swinging drill string support **600** is designed to support a hammer **605** or drill pipe **205** at the end of the drill string **150** when making a joint between the hammer **605** or drill pipe **205** and a drilling tool **215** in applications where the tower **110** is tilted and the hammer **605** or drill pipe **205** tends to sag out of alignment with the bit adaptor **235**, **535**.

(58) The side swinging drill string support **600** includes a mounting flange **605**, a piston **610**, a bracket **615**, and a roller **620**. The roller **620** is mounted to the bracket **615** via first support **617** and second support **619**. A spindle **618** extends between the supports **617**, **619** and through the roller **620** to enable the roller **620** to rotate with respect to the supports **617**, **619**. The roller **620** and bracket **615** combination are further rotationally coupled to the piston **610** via a pinned connection **625** between the piston **610** and the second support **619**.

(59) In the stowed configuration shown in FIG. **19A**, the piston **610** is in an extended position. When the piston **610** extends, it applies a pushing force to the second support **619** through the pinned connection **625**, thereby rotating the bracket **615** and the roller **620** into axial alignment with the mounting flange **605**. In the activated configuration shown in FIG. **19B**, the piston **610** is in a retracted position. When the piston **610** retracts, it applies a pulling force to the second support **619** through the pinned connection **625**, thereby rotating the bracket **615** and the roller **620** to a position that is perpendicular to the mounting flange **605**, as shown.

(60) FIG. **20A** and FIG. **20B** depict implementations of side swinging drill string supports **600**, **600'** in the activated configuration. The side swinging drill string supports **600**, **600'** may be identical in most respects, except for the rollers **620**, **620'**, which are sized to support different sized hammers **605**/drill pipes **205**.

(61) FIG. **21A** and FIG. **21B** depict the side swinging drill string support **600** in a representative

operational environment to provide support to a sagging hammer **605**/drill pipe **205** at the end of a drill string **150** and align it with the drilling tool **215**, such as the DTH tool **515**. As shown in FIG. **21A**, the side swinging drill string support **600** is in the stowed configuration, and the centerline **602** of the hammer **605** is sagging out of alignment with the centerline **517** of the DTH tool **515**. As shown in FIG. **21B**, when the side swinging drill string support **600** is in the activated configuration, the roller **620** of the side swinging drill string support **600** engages the hammer **605** and provides support to maintain alignment between the centerline **602** of the hammer **605** and the centerline **517** of the DTH tool **515**.

(62) In various embodiments, the side swinging drill string support **600**, **600'** may be configured to be operated manually, electrically, pneumatically, hydraulically, or by other methods. In some embodiments, the side swinging drill string support **600**, **600'** may be operatively coupled to the control system **155** such that the rig operator, or an automatic process in the control system **155**, determines when to move the side swinging drill string support **600**, **600'** from the stowed configuration to the activated configuration and vice versa.

(63) Referring now to FIG. **22A** through FIG. **25F**, in some embodiments, instead of the pivoting bit basket **535** and/or the side swinging drill string support **600**, **600'**, one or more alignment clamps may be installed within the enclosure **195** for the carousel assembly **230** of the drilling tool changer assembly **190**. The one or more alignment clamps may be used to align a drilling tool **215**, such as a DTH tool **515**, within a bit adaptor **235** and/or to align a drill pipe **205**/hammer **605** with the drilling tool **215**/DTH tool **515** to make the joint and avoid cross-threading.

(64) FIG. **22A** and FIG. **22B** depict side perspective views of an embodiment of an alignment clamp **700** shown in an open position and in a closed position, respectively. The alignment clamp **700** comprises a first arm **710**, a second arm **720**, an actuating piston **730**, and mounting connectors **736**, **738**. The first arm **710** and the second arm **720** are pivotally coupled via pin connections **732**, **734** to opposite ends of the actuating piston **730**. Each of the first arm **710** and the second arm **720** further comprises a pad system **715**, **725**, respectively, coupled to the end of the arm **710**, **720** opposite the pin connections **732**, **734**.

(65) Each of the pad systems **715**, **725** comprises a plurality of contact pads **714**, **724**, respectively, that are disposed substantially perpendicular to the plane of the arm **710**, **720** and positioned at different angles to one another. The contact pads **714** of the first arm **710** may be angularly oriented as mirror images to the contact pads **724** of the second arm **720**. As the alignment clamp **700** moves to the closed position shown in FIG. **22B**, the plurality of contact pads **714**, **724** partially surround and engage the DTH tool **515** to act as a wedge for alignment purposes, as further described herein.

(66) Each of the pad systems **715**, **725** further comprises a plurality of flared flanges **712**, **722**, respectively, each coupled to an upper end of a corresponding contact pad **714**, **724** and extending outwardly at an angle therefrom. The flared flanges **712** of the first arm **710** may be oriented as mirror images to the flared flanges **722** of the second arm **720**. When the alignment clamp **700** is in the closed position shown in FIG. **22B**, the plurality of flared pads **712**, **722** form a generally funnel shape operable to guide the hammer **605** for alignment purposes to allow the joint to be made, as further described herein.

(67) FIG. **23A** through FIG. **23C** depict the alignment clamp **700** of the present disclosure in a representative operational environment. As shown, the alignment clamp **700** may be installed, such as via mounting connectors **736**, **738**, within an enclosure **195** for a carousel assembly **230** comprising a plurality of bit adaptors **235**, one of which contains a DTH tool **515**. FIG. **23A** through FIG. **23C** depict a sequence of top plan views showing the alignment clamp **700** of the present disclosure moving the DTH tool **515** from a tilted position within a bit adaptor **235** to an aligned position within the bit adaptor **235** while the drilling tool changer assembly **190** is tilted to the exchange position shown in FIGS. **5-7**.

(68) As the enclosure **195** is tilted to the exchange position, the DTH tool **515** falls out of alignment with the bit adaptor **235**. FIG. **23A** shows the alignment clamp **700** of the present disclosure in an

open position, and the DTH tool **515** in a tilted position within the bit adaptor **235**. In this configuration, the actuating piston **730** of the alignment clamp **700** is extended, and the arms **710**, **720** of the alignment clamp **700** are pivoted open to allow rotation of the carousel assembly **230** without interference.

(69) FIG. **23B** shows the alignment clamp **700** with the arms **710**, **720** in a partially closed position, making contact with the DTH tool **515**. In more detail, as the actuating piston **730** of the alignment clamp **700** is retracted to move the alignment clamp **700** to a closed position, the arms **710**, **720** pivot about their respective pin connections **732**, **734** toward the DTH tool **515** until one of the contact pads **714** of the first arm **710** engages the DTH tool **515** at a first contact point **716** and the opposing one of the contact pads **724** of the second arm **720** engages the DTH tool **515** at a second contact point **726**. The contact points **716**, **726** are roughly 90 degrees offset from one another.

(70) FIG. **23C** shows the alignment clamp **700** of the present disclosure in a closed position around the DTH tool **515**, which is now aligned with the bit adaptor **235**. In more detail, as the actuating piston **730** continues retracting to close the alignment clamp **700**, the arms **710**, **720** continue to pivot inwardly about their respective pin connections **732**, **734**, and the contact pads **714**, **724** act as a wedge to push the DTH tool **515**. As the alignment clamp **700** tightens its grip, the DTH tool **515** is drawn towards a substantially centered position relative to the contact pads **714**, **724**. In the aligned position shown in FIG. **23C**, the contact pads **714**, **724** engage the DTH tool **515** at four points of contact **716**, **718**, **726**, **728** that are spaced apart by roughly 90 degrees.

(71) FIG. **24** depicts a side perspective view of the assembly shown in FIG. **23C**, with the alignment clamp **700** of the present disclosure shown in the closed position and with the DTH tool **515** aligned with the bit adaptor **235**. With the contact pads **714**, **724** holding the DTH tool **515** in alignment, the flared flanges **712**, **722** form a generally funnel shape that will help guide the hammer **605** to maintain alignment between the centerline **602** of the hammer **605** and the centerline **517** of the DTH tool **515** to facilitate making the joint without cross-threading.

(72) In some embodiments, multiple alignment clamps may be used instead of a single alignment clamp. In such embodiments, a lower alignment clamp may align the DTH tool **515** with the bit adaptor **235**, and an upper alignment clamp may align the hammer **605** with the DTH tool **515** to avoid cross threading when the joint is made.

(73) FIG. **25A** through FIG. **25F** depict an assembly using multiple alignment clamps of the present disclosure in a representative operational environment. In one embodiment, a lower alignment clamp **700'** is provided that is structurally similar to the alignment clamp **700** of FIG. **22A** through FIG. **24**, except the lower alignment clamp **700'** does not include the flared flanges **712**, **722** at the top of the contact pads **714**, **724**. An upper alignment clamp **700** is also provided that includes the flared flanges **712**, **722** at the top of the contact pads **714**, **724**.

(74) FIG. **25A** through FIG. **25F** depict a sequence of side perspective views showing a tower **110** carrying a drill string **150** with a hammer **605** at an end thereof and a drilling tool changer assembly **190** tilted to the exchange position. The drilling tool changer assembly **190** includes an enclosure **195** that houses the lower and upper alignment clamps **700'**, **700**, and a carousel assembly **230** with a plurality of bit adaptors **235**, one of which holds a DTH tool **515**.

(75) FIG. **25A** shows the lower and upper alignment clamps **700'**, **700** in open positions, respectively, and the top heavy DTH tool **515** tilted within the bit adaptor **235**. FIG. **25B** shows the lower alignment clamp **700'** in the closed position around the DTH tool **515** to thereby align the DTH tool **515** within the bit adaptor **235** to allow for threading to the hammer **605**.

(76) Once the DTH tool **515** has been aligned, the drill string **150** can then move the hammer **605** toward the DTH tool **515**. During this movement, as shown in FIG. **25C**, the upper alignment clamp **700** is closed, and the flared flanges **712**, **722** of the upper alignment clamp **700** guide the hammer **605** into alignment with the DTH tool **515** to avoid cross threading. FIG. **25D** shows the lower and upper alignment clamps **700'**, **700** closed to maintain alignment as the hammer **605** is threaded onto the DTH tool **515**.

(77) As shown in FIG. 25E, once the joint is made, the lower and upper alignment clamps **700'**, **700** can be retracted to the open position, and as shown in FIG. 25F, the drill string **150**, hammer **605**, and DTH tool **515** are retracted out of the enclosure **195**.

(78) In various embodiments, the alignment clamps **700'**, **700** may be configured to be operated manually, electrically, pneumatically, hydraulically, or by other methods. In some embodiments, the alignment clamps **700'**, **700** may be operatively coupled to the control system **155** such that the rig operator, or an automatic process in the control system **155**, determines when to move each of the alignment clamps **700'**, **700** from the open position to the closed position and vice versa.

(79) None of the description in this application should be read as implying that any particular element, step, or function is an essential element which must be included in the claim scope; the scope of patented subject matter is defined only by the allowed claims. Moreover, none of these claims are intended to invoke 35 U.S.C. Section 112(f) unless the exact words “means for” are used, followed by a gerund. The claims as filed are intended to be as comprehensive as possible, and no subject matter is intentionally relinquished, dedicated, or abandoned.

Claims

1. An apparatus for changing a drilling tool for a drilling rig, wherein the drilling rig comprises a drill tower supporting a drill pipe; the apparatus for changing a drilling tool comprising: a drilling tool changer assembly moveable between a storage position and an exchange position; the drilling tool changer assembly supporting a rotatable carousel assembly; the rotatable carousel assembly having a plurality of bit adaptors for holding drilling tools; the rotatable carousel assembly selectively rotatable to bring a selected drilling tool into coaxial alignment with the drill pipe when the drilling tool changer assembly supporting the rotatable carousel assembly is moved into the exchange position; and means for aligning the selected drilling tool within a bit adaptor holding the selected drilling tool.
2. The apparatus for changing a drilling tool of claim 1, further comprising: means for aligning the drill pipe with the selected drilling tool.
3. The apparatus for changing a drilling tool of claim 1, wherein the selected drilling tool is a down-the-hole (“DTH”) tool.
4. The apparatus for changing a drilling tool of claim 3, further comprising: a hammer coupled to the drill pipe.
5. The apparatus for changing a drilling tool of claim 4, further comprising: means for aligning the hammer with the DTH tool.
6. The apparatus for changing a drilling tool of claim 1, where the rotatable carousel assembly is further selectively rotatable to bring a selected bit adaptor of the plurality of bit adaptors into a pre-determined angular position for pick-up or drop-off of a drilling tool in the rotatable carousel assembly.
7. The apparatus for changing a drilling tool of claim 1, where the bit adaptors further comprise at least one nub for holding a drilling tool against a torque imparted to the drilling tool.
8. The apparatus for changing a drilling tool of claim 1, where the drilling tool changer assembly is pivotably hinged to bring the rotatable carousel assembly into coaxial alignment with a drill pipe supported by the drill tower.
9. The apparatus for changing a drilling tool of claim 1, where the bit adaptors are differently sized to accommodate differently-sized drilling tools.
10. The apparatus for changing a drilling tool of claim 1, further comprising a control system for controlling operations of the drilling tool changer assembly.
11. The apparatus for changing a drilling tool of claim 10, further comprising a first actuator for selectively rotating the rotatable carousel assembly under command of the control system.
12. The apparatus for changing a drilling tool of claim 10, further comprising a second actuator for

selectively moving the drilling tool changer assembly between the storage position and the exchange position under command of the control system.

13. The apparatus for changing a drilling tool of claim 1, where the drilling tool changer assembly further comprises a sensor for identifying the one or more drilling tools located in the bit adaptors.

14. The apparatus for changing a drilling tool of claim 1, where the drilling tool changer assembly further comprises a sensor for detecting an angular position of the rotatable carousel assembly.

15. The apparatus for changing a drilling tool of claim 1, wherein the drilling tool changer assembly further comprises a detector for detecting the presence or absence of a drilling tool in one of the plurality of bit adaptors.

16. The apparatus for changing a drilling tool of claim 1, where the drilling tool changer assembly further comprises an enclosure; the enclosure enclosing the rotatable carousel assembly.

17. The apparatus for changing a drilling tool of claim 16, where the drilling tool changer assembly further comprises a lid and a third actuator; the lid covering the enclosure; and, wherein the lid is deployed between an open position and a closed position by action of the third actuator.
